

## 1. Analisis Set A (Ronde 1-10)

### Data Input:

- $TP = 45$  (Deteksi Benar)
- $FN = 165$  (Gagal Deteksi)
- $FP = 0$  (Salah Alert)
- $TN = 140$  (Normal Benar)
- Total Sampel ( $N$ ) = 350

### Perhitungan:

- **Akurasi (Accuracy):**

$$Accuracy = \frac{45 + 140}{350} = \frac{185}{350} = 52.86\%$$

- **Presisi (Precision):**

$$Precision = \frac{45}{45 + 0} \times 100\% = 100\%$$

- **Recall (Detection Rate):**

$$Recall = \frac{45}{45 + 165} = \frac{45}{210} = 21.43\%$$

- **F1-Score:**

$$F1 = 2 \times \frac{1 \times 0.2143}{1 \times 0.2143} = 35.29\%$$

## 2. Analisis Set B (Ronde 11-20)

### Data Input:

- $TP = 50$  (Deteksi Benar)
- $FN = 160$  (Gagal Deteksi)
- $FP = 0$  (Salah Alert)
- $TN = 140$  (Normal Benar)
- Total Sampel ( $N$ ) = **350**

### Perhitungan:

- **Akurasi (Accuracy):**

$$Accuracy = \frac{50 + 140}{350} = \frac{190}{350} = \mathbf{54.29\%}$$

- **Presisi (Precision):**

$$Precision = \frac{50}{50 + 0} \times 100\% = \mathbf{100\%}$$

- **Recall (Detection Rate):**

$$Recall = \frac{50}{50 + 160} = \frac{50}{210} = \mathbf{23.81\%}$$

- **F1-Score:**

$$F1 = 2 \times \frac{1 \times 0.2381}{1 \times 0.2381} = \mathbf{38.46\%}$$

### 3. Analisis Set C (Ronde 21-30)

#### Data Input:

- $TP = 52$  (Deteksi Benar)
- $FN = 158$  (Gagal Deteksi)
- $FP = 0$  (Salah Alert)
- $TN = 140$  (Normal Benar)
- Total Sampel ( $N$ ) = **350**

#### Perhitungan:

- **Akurasi (Accuracy):**

$$Accuracy = \frac{52 + 140}{350} = \frac{192}{350} = \mathbf{54.86\%}$$

- **Presisi (Precision):**

$$Precision = \frac{52}{52 + 0} \times 100\% = \mathbf{100\%}$$

- **Recall (Detection Rate):**

$$Recall = \frac{52}{52 + 158} = \frac{52}{210} = \mathbf{24.76\%}$$

- **F1-Score:**

$$F1 = 2 \times \frac{1 \times 0.2476}{1 \times 0.2476} = \mathbf{39.69\%}$$

### Rekapitulasi Data Total (Ronde 1-30)

Kita menjumlahkan hasil dari ketiga set sebelumnya:

- **Total True Positive (TP):**  $45 + 50 + 52 = 147$
- **Total False Negative (FN):**  $165 + 160 + 158 = 483$
- **Total False Positive (FP):**  $0 + 0 + 0 = 0$
- **Total True Negative (TN):**  $140 + 140 + 140 = 420$

**Total Sampel Data (N):**  $147 + 483 + 0 + 420 = 1.050$  Data Point.

### Perhitungan Metrik Evaluasi & Rumus

#### A. Akurasi (Accuracy)

Mengukur ketepatan sistem secara keseluruhan (deteksi benar + normal benar).

$$Accuracy = \frac{147 + 420}{1.050} \times 100\% = \frac{567}{1.050} = 54.00\%$$

#### B. Presisi (Precision)

Mengukur kepercayaan terhadap alert yang muncul. Karena tidak ada *False Alarm*, nilai ini sempurna.

$$Precision = \frac{147}{147 + 0} \times 100\% = 100\%$$

#### C. Recall (Detection Rate)

Mengukur persentase serangan yang berhasil dideteksi dari total serangan yang dilancarkan.

$$Recall = \frac{147}{147 + 483} \times 100\% = \frac{147}{630} = 23.33\%$$

#### D. F1-Score

Nilai rata-rata harmonik untuk menyeimbangkan Presisi dan Recall.

$$F1 = 2 \times \frac{1 \times 0.2333}{1 \times 0.2333} = 37.84\%$$